

INCIDENT ANALYSIS REPORT

# SOC Challenge

## Network Traffic Analysis

*Suspected Info-Stealer Malware - tcpdump\_challenge.pcap*

**⚠ SEVERITY: HIGH - Credential Exfiltration & Malware Activity Confirmed**

**Organization:** Astley Financial

**Analyst:** Michael Leramo

**Date:** 24 May 2026

**Classification:** CONFIDENTIAL

**Assessment:** SOC Challenge

**Evidence File:** tcpdump\_challenge.pcap

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## 1. Executive Summary

A SOC alert was triggered at Astley Financial after an endpoint exhibited abnormal behavioural patterns consistent with info-stealer malware activity. This report documents the analysis of the captured network traffic (tcpdump\_challenge.pcap) to identify the scope of compromise, confirm indicators of infection, and present actionable findings.

Analysis confirmed credential theft via HTTP POST requests, lateral movement over a file-sharing protocol, and communication with a known malware command-and-control infrastructure.

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## 2. Incident Overview

|                        |                                                      |
|------------------------|------------------------------------------------------|
| <b>Organisation</b>    | Astley Financial                                     |
| <b>Alert Type</b>      | Abnormal endpoint behaviour / potential info-stealer |
| <b>Evidence File</b>   | tcpdump_challenge.pcap                               |
| <b>Analysis Method</b> | tcpdump + Wireshark                                  |
| <b>Analyst</b>         | Michael Leramo                                       |
| <b>Date</b>            | 24 May 2026                                          |

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## 3. Analysis Methodology

The PCAP was opened in Wireshark and inspected with targeted tcpdump filters. Analysis progressed through four phases:

- Traffic volume enumeration - total packet count and protocol distribution
- ICMP reconnaissance - identifying ping targets and external ASN lookup
- HTTP traffic analysis - extracting POST payloads, credentials, and User-Agent strings
- File-sharing protocol inspection - recovering authentication credentials and retrieved files

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## 4. Findings

### 4.1 Packet Capture Statistics

| Metric                | Value    |
|-----------------------|----------|
| Total Packets         | 1344     |
| ICMP Packets          | 132      |
| HTTP POST Requests    | 1        |
| Secondary Active Port | 21 (FTP) |

I loaded the PCAP in Wireshark and used the Statistics > Protocol Hierarchy view alongside targeted display filters to confirm the above counts. The packet count was also verified with tcpdump.

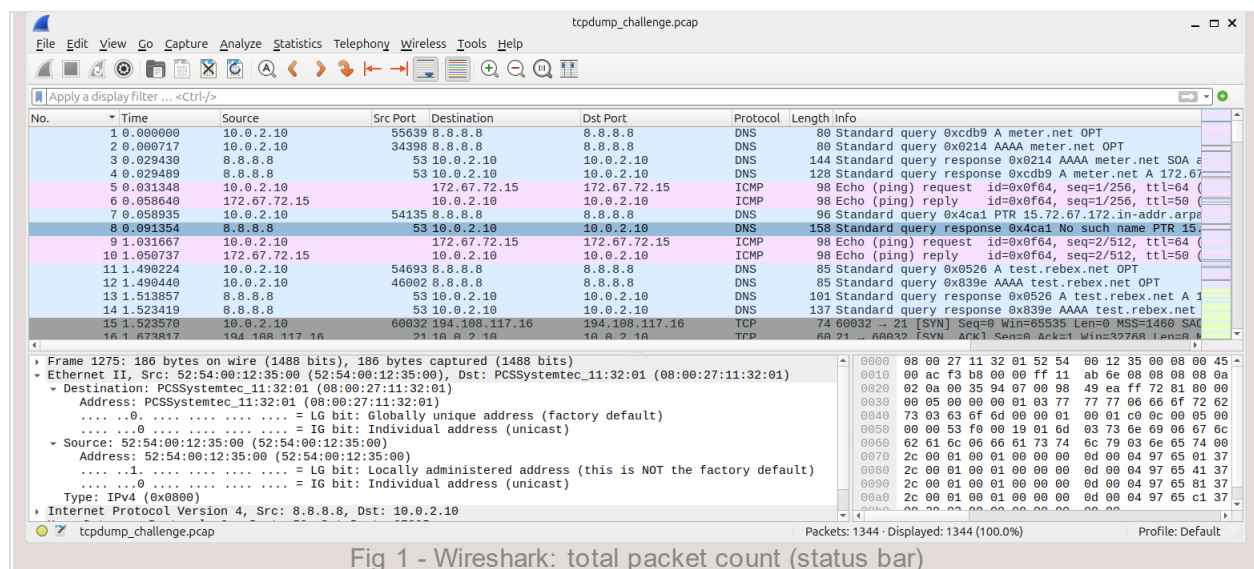


Fig 1 - Wireshark: total packet count (status bar)

## 4.2 ICMP - Ping Reconnaissance

The endpoint was observed sending ICMP echo requests to an external IP address. I identified the destination and performed an ASN lookup to determine ownership.

```
tcpdump -r tcpdump_challenge.pcap icmp -n
```

| Item                | Value                |
|---------------------|----------------------|
| Ping Destination IP | 149.154.167.99       |
| ASN                 | AS13335 (Cloudflare) |

An endpoint pinging external infrastructure at regular intervals can indicate a connectivity check by malware before it initiates exfiltration, a behaviour consistent with the info-stealer hypothesis.

```
bigmyke@cyber-nonja:~/Downloads/SOC-FILE-RESOURCES/02_Network_Security/01_tcpdump/Challenge$ tcpdump -r tcpdump_challenge.pcap -n icmp -c 5
reading from file tcpdump_challenge.pcap, link-type EN10MB (Ethernet), snapshot length 262144
15:53:51.081222 IP 10.0.2.10 > 172.67.72.15: ICMP echo request, id 3940, seq 1, length 64
15:53:51.108514 IP 172.67.72.15 > 10.0.2.10: ICMP echo reply, id 3940, seq 1, length 64
15:53:52.081541 IP 10.0.2.10 > 172.67.72.15: ICMP echo request, id 3940, seq 2, length 64
15:53:52.100611 IP 172.67.72.15 > 10.0.2.10: ICMP echo reply, id 3940, seq 2, length 64
15:53:53.081802 IP 10.0.2.10 > 172.67.72.15: ICMP echo request, id 3940, seq 3, length 64
```

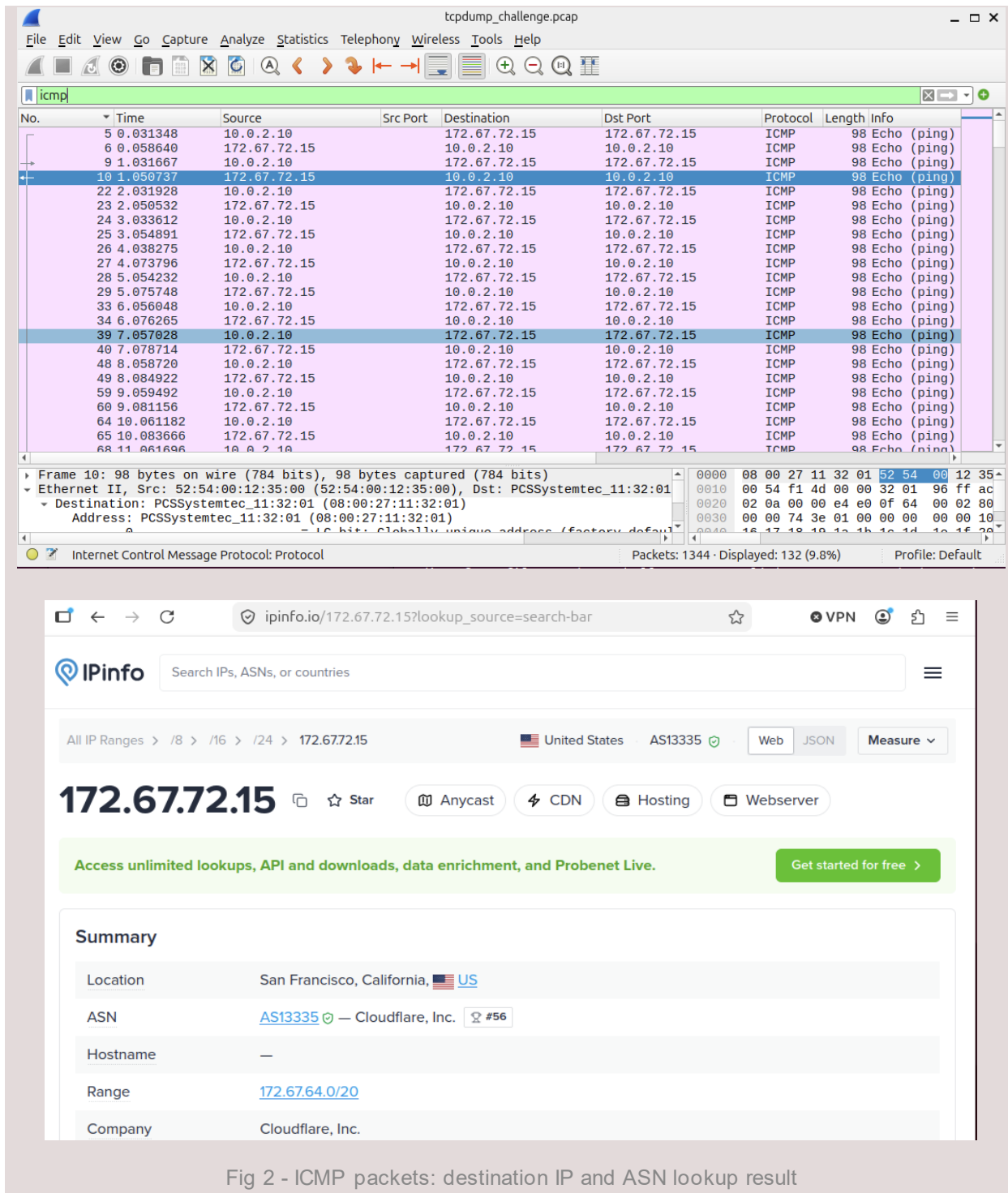


Fig 2 - ICMP packets: destination IP and ASN lookup result

## 4.3 HTTP Traffic - Credential Exfiltration

### 4.3.1 POST Requests

HTTP POST requests were examined to identify credential data sent in plaintext. Wireshark's Follow TCP Stream feature was used to inspect the full payload of each POST.

```
http.request.method == "POST"
```

Password recovered from HTTP POST payload:

| Field           | Value         |
|-----------------|---------------|
| HTTP POST Count | 1             |
| Password Found  | ilovecats9102 |

The transmission of credentials in cleartext over HTTP port 80 indicates the malware was exfiltrating harvested credentials with no encryption, a hallmark of low-sophistication but highly effective info-stealer operation.

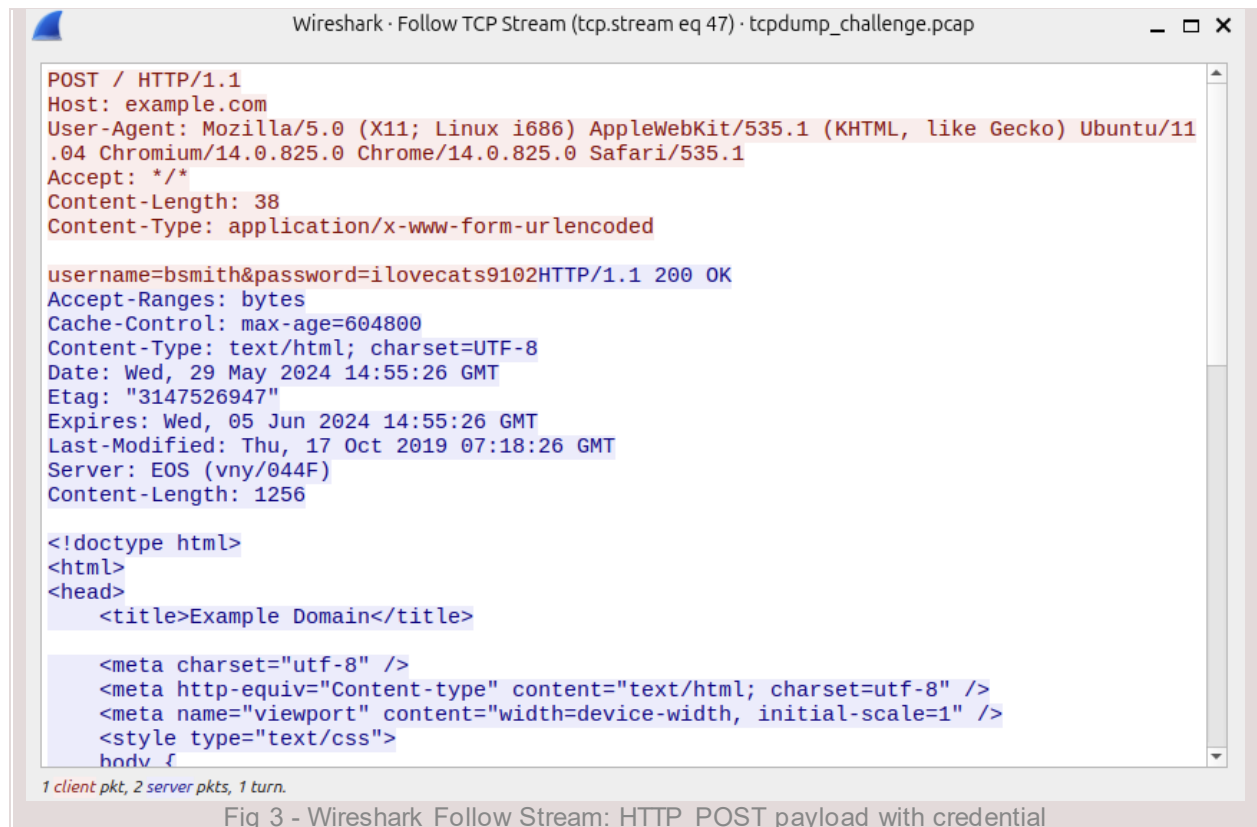


Fig 3 - Wireshark Follow Stream: HTTP POST payload with credential

#### 4.3.2 User-Agent String Analysis

The HTTP requests carried an anomalous User-Agent string not typical of any standard browser or update agent. I extracted this and cross-referenced with open threat intelligence sources.

```
http.user_agent
```

| Field              | Value                         |
|--------------------|-------------------------------|
| User-Agent String  | TeslaBrowser/5.5              |
| Identified Malware | Lumma Stealer                 |
| C2 URL (defanged)  | hxxps[:]//t[.]me/+zz0192lskaa |

Matching the User-Agent to known malware families is a standard threat intelligence technique. The identified malware family helped confirm the nature of the infection and is consistent with the credential-stealing behaviour observed in the POST payloads.

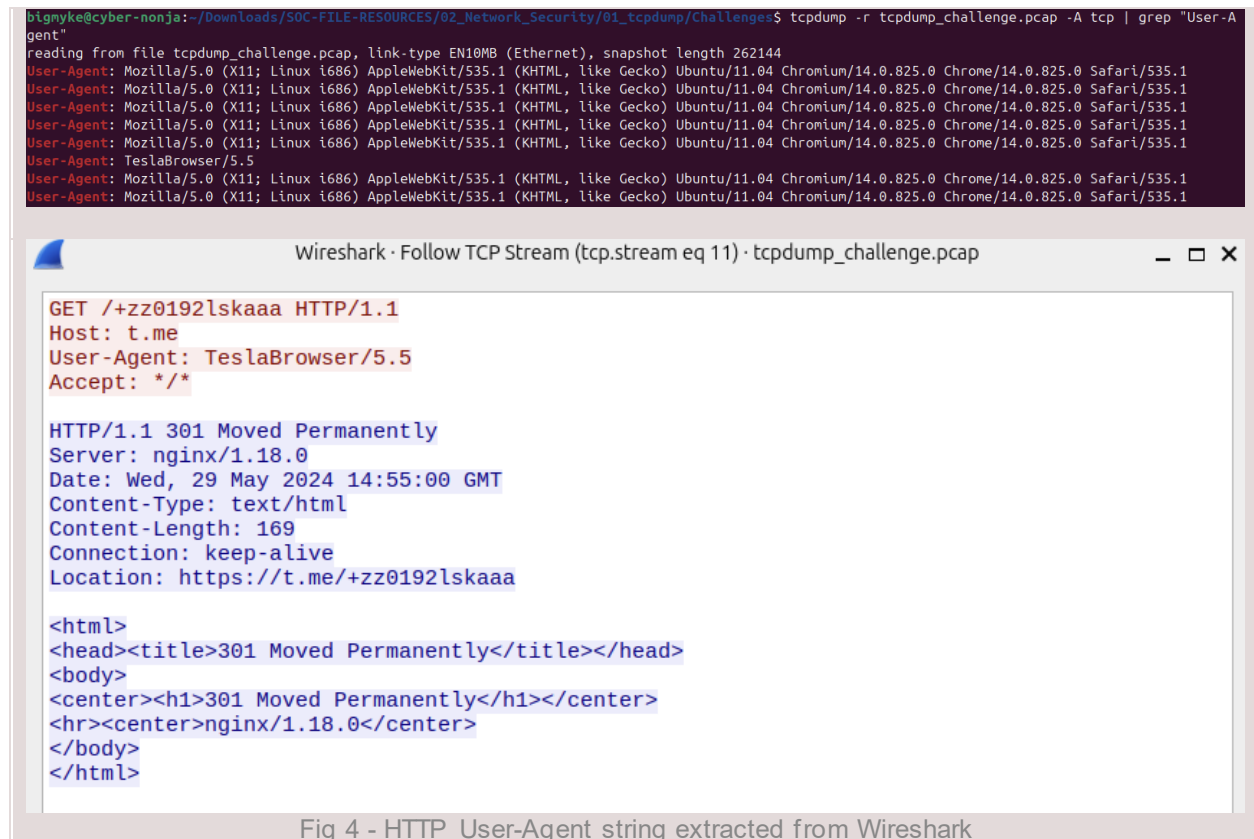


Fig 4 - HTTP User-Agent string extracted from Wireshark

#### 4.4 File Sharing - Lateral Movement / Data Staging

Traffic on a secondary well-known port was identified as a file-sharing protocol. I extracted authentication exchanges to recover valid credentials and identified the file that was retrieved.

| Item                    | Value         |
|-------------------------|---------------|
| Protocol Port           | 21 / FTP      |
| Credentials (user:pass) | demo:password |
| File Retrieved          | readme.txt    |

Valid credentials recovered from the file-sharing session suggest either the malware had access to stored credentials, or the credentials were harvested prior to this session and reused. The retrieved file may contain additional payloads or sensitive data.

| No. | Time      | Source          | Src Port | Destination    | Dst Port       | Protocol | Length | Info                                                            |
|-----|-----------|-----------------|----------|----------------|----------------|----------|--------|-----------------------------------------------------------------|
| 153 | 24.958131 | 194.108.117.100 | 21       | 10.0.2.10      | 10.0.2.10      | FTP      | 158    | Response: 220-Welcome to test.rebex.net!                        |
| 155 | 24.958295 | 194.108.117.100 | 21       | 10.0.2.10      | 10.0.2.10      | FTP      | 121    | Response: 220 If you don't have an account, log in as 'anonymou |
| 159 | 25.738383 | 10.0.2.10       | 40124    | 194.108.117.16 | 194.108.117.16 | FTP      | 65     | Request: USER demo                                              |
| 161 | 25.870831 | 194.108.117.100 | 21       | 10.0.2.10      | 10.0.2.10      | FTP      | 130    | Response: 331 Anonymous login OK, send your complete email addr |
| 167 | 27.590260 | 10.0.2.10       | 40124    | 194.108.117.16 | 194.108.117.16 | FTP      | 69     | Request: PASS password                                          |
| 168 | 27.723027 | 194.108.117.100 | 21       | 10.0.2.10      | 10.0.2.10      | FTP      | 82     | Response: 230 User 'demo' logged in.                            |

Fig 5 - File sharing session: authentication and file retrieval

## 5. Indicators of Compromise (IOCs)

| IOC Type                | Value                           | Context                                           |
|-------------------------|---------------------------------|---------------------------------------------------|
| Destination IP          | 149.154.167.99                  | Endpoint pinged externally — possible C2 check    |
| Credential (HTTP)       | ilovecats9102                   | Password exfiltrated in plaintext via HTTP POST   |
| Credential (File Share) | demo:password                   | Valid credentials used in file-sharing session    |
| Malware Family          | Lumma Stealer                   | Identified via User-Agent string in HTTP requests |
| C2 URL (defanged)       | hxxps[:]//[t[.]me/+zz0192lksaaa | Endpoint attempted connection to this URL         |

## 6. Conclusion & Recommendations

The packet capture analysis confirmed that the endpoint was infected with what appears to be an info-stealer malware variant. Three distinct threat behaviours were observed: external host connectivity checks via ICMP, credential exfiltration over unencrypted HTTP, and authenticated access to an internal file-sharing server.

My intelligence background has shaped how I read network activity — I look for behavioural sequences, not just individual events. The ICMP pings, followed by HTTP POST credential dumps, followed by file-sharing access, form a coherent attack chain: check connectivity, exfiltrate credentials, stage or retrieve files.

### Recommended Actions

- Immediately isolate the affected endpoint from the network
- Reset all credentials identified in the PCAP, both HTTP and file-sharing
- Block the C2 URL at the perimeter firewall and DNS level
- Scan the environment for lateral movement using the recovered file-sharing credentials as a pivot point
- Enforce HTTPS across all internal services to prevent future plaintext credential capture

- Submit the malware User-Agent string to threat intel feeds for broader detection coverage

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## 7. Analyst Notes

This analysis was conducted as part of a practical SOC assessment scenario. All findings are based solely on the provided PCAP. The scenario mirrors real-world triage tasks a junior SOC analyst would perform in a L1 role.

Tools used: tcpdump (command-line filtering), Wireshark (traffic inspection and stream following), ipinfo.io / ARIN WHOIS (ASN lookup).